



Glyphosate fate in soils when arriving in plant residues



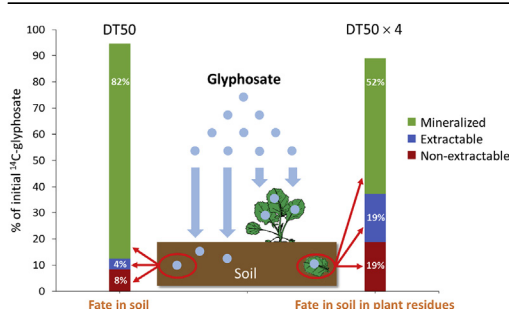
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HIGHLIGHTS

- Better insight into the fate of pesticide returned to soils via plant residues is needed.
- Incorporation into oilseed rape residues hampered the mineralization of glyphosate in soils.
- Incorporation into plant residues increased the amounts of glyphosate and of its metabolite in soils.
- The trapping of pesticides into plant materials increases their persistence in soils.

GRAPHICAL ABSTRACT



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ABSTRACT

A significant fraction of pesticides sprayed on crops may be returned to soils via plant residues, but its fate has been little documented. The objective of this work was to study the fate of glyphosate associated to plants residues. Oilseed rape was used as model plant using two lines: a glyphosate-tolerant (GT) line and a non-GT one, considered as a crucifer weed. The effects of different fragmentation degrees and placements in soil of plant residues were tested. A control was set up by spraying glyphosate directly on the soil. The mineralization of glyphosate in soil was slower when incorporated into plant residues, and the amounts of extractable and non-extractable glyphosate residues increased. Glyphosate availability for mineralization increased when the size of plant residues decreased, and as the distribution of plant residues in soil was more homogeneous. After 80 days of soil incubation, extractable ^{14}C -residues mostly involved one metabolite of glyphosate (AMPA) but up to 2.6% of initial ^{14}C was still extracted from undecayed leaves as glyphosate. Thus, the trapping of herbicides in plant materials provided a protection against degradation, and crops residues returns may increase the persistence of glyphosate in soils. This pattern appeared more pronounced for GT crops, which accumulated more non-degraded glyphosate in their tissues.

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1. Introduction

A fraction of pesticides applied to crops (especially for foliar pesticides) is intercepted and absorbed by the leaves of weeds and/

or crops. Plant material containing pesticides is returned to the soil during the plant cycle (via leaf senescence) or after harvest (as crop residues). The amounts of pesticide thus returned to soil with plant can be significant (Doublet et al., 2009; Von Wiren-Lehr et al., 1997), in particular with soil conservation practices because they lead to a continuous coverage of the soil surface by plant residues (Beare et al., 1993; Guérif et al., 2001). Although these practices have been known for a long time, their use was strongly increased with the introduction of glyphosate-tolerant (GT) crops, which are

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among the most cultivated genetically modified crops in the world (Cerdeira and Duke, 2006; Jacobsen et al., 2013; Locke et al., 2008). Glyphosate, a broad spectrum post-emergence herbicide, was already applied for weed control in conservation tillage systems (Dorn et al., 2013). Its use in GT crops will lead to additional inputs to soils through plant residues, especially since glyphosate is hardly degraded by most GT plants (Cerdeira and Duke, 2006; Dill, 2005; Locke et al., 2008; Nandula et al., 1999; Nandula et al., 2007). Once glyphosate has been absorbed by plants, its release into the environment depends on the decay rate of dead plant material, since glyphosate will remain sequestered in plant tissue until the plant dies and starts decomposing (Locke et al., 2008).

While the fate of glyphosate in soils is well-known (e.g. Cerdeira and Duke, 2006; Helander et al., 2012; Mamy et al., 2005), the soil fate of glyphosate absorbed in plant material (and that of other pesticides in general) has been little investigated in the literature. A soil incubation of glyphosate associated with soybean cells showed the fate of this pesticide to differ from a direct application on bare soil: glyphosate mineralization involved a lag phase and the fraction of non-extractable residues (NER) increased (Von Wiren-Lehr et al., 1997). Doublet et al. (2009) showed that the soil fate of glyphosate in aerial parts of oilseed rape and/or maize was different from that of glyphosate being directly applied to soils. Soil mineralization of glyphosate in crops decreased, and amounts of ^{14}C -extractable residues, mainly composed by the metabolite aminomethylphosphonic acid (AMPA), and NER increased. The fate of glyphosate was influenced by the type of plant compartment in which the herbicide was absorbed, because of differences in herbicide bioavailability and plant compartments biodegradability. However these two pieces of work do not consider the effects of the agricultural practices on the fate of plant residues and consequently on the fate of the associated glyphosate.

Under conventional tillage practices, crop residues are partially or totally incorporated into the soil, while in no-tillage management systems, the soil is not plowed and crop residues accumulate on the soil surface as a mulch layer (Beare et al., 1993; Guérif et al., 2001). Tillage directly affects plant residues fragmentation and distribution, and indirectly affects the environmental conditions which drive residues decomposition, such as moisture content and temperature (Angers and Recous, 1997; Guérif et al., 2001). Ground residue materials are more susceptible to microbial attack than intact plant parts due to a better soil-residue contact; however, fine particles are also more likely to be protected against decomposition through physical protection by clay and other particles (Angers and Recous, 1997; Beare et al., 1993; Iqbal et al., 2014; Sørensen et al., 1996). In addition, the placement of crop residues in soil dramatically affects the dynamics of residue decomposition, the latter being faster for buried than surface residues (Coppens et al., 2006; Douglas et al., 1980; Guérif et al., 2001). Soil water content, temperature and nutrient proximity are among the most important variables that are affected by plant residues placement (Beare et al., 1993). To the best of our knowledge, the effect of crop residues form and of their placement on the soil fate of pesticide contained in plants is not known.

In this context, the objective of this work was to study the fate in soil of glyphosate in glyphosate-tolerant and non-tolerant oilseed rape, as a function of the size of plant residues and their placement in soils. Experiments involved a preliminary step of ^{14}C -glyphosate absorption on leaves, followed by laboratory incubations of soil samples combined with leaves using different placement and fragmentation procedures. Oilseed rape was selected because it is one of the most cultivated GT crops in the world (Jacobsen et al., 2013) and also because it may be considered as a model for crucifer weeds.

2. Materials and methods

2.1. Herbicide

[Methyl- ^{14}C]glyphosate (N-(phosphonomethyl)glycine) was purchased from Sigma Chemicals (81 MBq mmol $^{-1}$, 99.2% purity). Water solution of labeled glyphosate was prepared at 372 mg L $^{-1}$ (containing 166 MBq L $^{-1}$) in order to add 18 μg of glyphosate (0.08 MBq) on each treated leaf (Grangeot et al., 2006; Lutman et al., 2008; Stanton et al., 2010; see section 2.4. Incubation procedure).

2.2. Soil

Soil samples were taken from the top layer (0–10 cm) of a French experimental site (Dijon, Burgundy), immediately placed in a cooler and taken to the laboratory where they were passed through a 3 mm sieve, removing visible organic residues by hand (Angers and Recous, 1997), and stored at 4 °C for 8 days before use. The soil is a clay-loam calcareous Cambisol (IUSS, 2015) with (% of dry soil): 37.7 of clay, 29.6 of silt, 15.2 of sand, 16.7 of CaCO_3 , 1.63 of organic carbon, and pH in water of 8.2.

2.3. Plant material

Glyphosate-tolerant (Roundup Ready $^{\text{®}}$) and non-tolerant varieties of oilseed rape (*Brassica napus* L.) were sampled in French agricultural experimental sites, both at the 4-leaf development stage. In this work, non-GT oilseed rape is considered as a representative of crucifer weeds, which commonly occur in oilseed rape fields (Légère, 2005). Direct comparison between GT and non-GT oilseed rape crops is not possible as the two oilseed rapes are not issued from the same genetic line.

Twenty-four hours following harvest of oilseed rape, the youngest leaves were cut off and treated in laboratory with ten 5 μL droplets of ^{14}C -glyphosate solution using a 25 μL micro syringe (Hamilton Co, Alltech) (Chamel et al., 1991). Preliminary results showed that there was no difference in glyphosate absorption by leaves between a whole oilseed rape plant and an isolated oilseed rape leaf (Mamy, 2004). The amounts of herbicide applied (18 μg /leaf) corresponded to a glyphosate treatment of 540 g ha $^{-1}$ in an application volume of 150 L ha $^{-1}$ consistent with the dose of glyphosate usually used for weed control in GT oilseed rape (Grangeot et al., 2006; Lutman et al., 2008; Stanton et al., 2010). The treated leaves were selected for the incubation experiments as they contain the highest amounts of glyphosate among all plants parts, and therefore represent the main contribution of glyphosate input to soil through crop residues (Doublet et al., 2009; Nandula et al., 1999).

Eight days after glyphosate application, one half of the treated leaves were washed with 10 mL of ultrapure water (Millipore) (Doublet et al., 2009; Nandula et al., 1999). The amounts of absorbed herbicide in washed leaves were estimated by the difference between radioactivity contained in wash solution, determined by liquid scintillation counting (see section 2.5. Chemical analysis), and the applied radioactivity. The washed leaves allowed incubation of the amount of herbicide that was absorbed in the leaves, while the non-washed leaves represented the maximum amount of herbicide that can be returned to the soil through leaves.

Treated leaves were prepared to study different conditions of plant materials when they reach the soil (Table 1). The effect of the size of plant residues was studied comparing entire leaves, leaves fragments of 3-mm size obtained after fractionation with scalpel, or crushed leaves finely ground in a glass mortar. To study the effect of plant residues placement in soil, leaves will be put on the soil

Table 1

Conditions of soil incubation of glyphosate residues in glyphosate-tolerant and non-tolerant oilseed rape.

Leaf fragmentation	Soil placement
Entire	Surface
Entire	Middle
3 mm	Homogeneous
Crushed	Homogeneous
Crushed, in 20 μm porosity polyamide bag	Middle

surface, placed in the middle of the soil sample or mixed homogeneously with the soil, depending on their fragmentation (Table 1). In addition, we added a treatment where a crushed leaf was placed in a polyamide bag (Nytrel-Ti) of 20 μm porosity (average section of 5 cm^2) (Benoit et al., 1999). This last setup enabled us to cleanly recover the leaves samples to study the transfer of radioactivity from leaves to soil.

2.4. Incubation procedure

Laboratory incubations of treated leaves were done with unit soil samples of 10 g dry soil, to obtain an average plant material:soil ratio of 3 g kg^{-1} (3.4 ± 1.2 g kg^{-1}). This ratio was deemed representative of crop residues return rates to soils (Johnson et al., 2006). Controls were set up by direct application of 50 μL of glyphosate solution on 10 g dry soil.

The soil samples, with or without plant materials, were placed in 500 mL hermetically stoppered jars and incubated at 28 ± 1 °C in the dark for 80 days. Water was added to reach a soil moisture corresponding to a pF of 3. During incubation, soil water content was adjusted periodically by weighing each jar and adding the required amount of water. The jars contained a vial with 5 mL of 2 M NaOH to trap the $^{14}\text{CO}_2$ and the total CO_2 evolved, and a vial with 10 mL of water to keep the relative humidity constant. NaOH traps were removed and replaced 1, 3, 7, 14, 21, 28, 42, 61 and 80 days after treatment. Three replicates were done for each modality and for the control so that 63 jars were incubated: 3 for control, 15 for GT oilseed rape washed leaves (3 jars for each of the 5 modalities of leaf fractionation and placement in soil, Table 1), and similarly for non-washed GT oilseed rape leaves, and washed and non-washed non-GT oilseed rape leaves.

At the end of the incubation, soil samples were extracted once with 50 mL of 0.01 M CaCl_2 water solution for 24 h, then three times (24–4–24 h) with 50 mL of 0.54 M NH_4OH (Aubin and Smith, 1992). Entire leaf left on the soil surface was removed from the soil and extracted separately with 10 mL of the same solvents (Rampoldi et al., 2011). Non-extractable residues in soil and leaves corresponded to the radioactivity remaining after the four extractions. The leaf samples in the polyamide bags were combusted but not extracted (to avoid loss of leaf fragments during extraction), so that bags contained both extractable and non-extractable ^{14}C .

2.5. Chemical analysis

Glyphosate mineralization was determined by measuring $^{14}\text{CO}_2$ in 1 mL of NaOH traps by liquid scintillation counting (Tri-carb 2100 TR counter, Packard Instruments), and total soil respiration (total CO_2) was measured in the remaining 4 mL of NaOH traps using a colorimetric method with a continuous flow analyzer (Skalar).

Total radioactivity content of CaCl_2 and NH_4OH extracts was measured by liquid scintillation counting. Extracts were analyzed by HPLC after they were concentrated by evaporation (Büchi) under vacuum, filtered through a syringe-regenerated cellulose filter

(0.45 μm , Alltech), and subsequently acidified to pH 2 with H_3PO_4 . HPLC analyses were performed with a Waters chromatography appliance (System controller 610, Autosampler 717) coupled with a radioactive flow detector (Flo-one A-500, Packard-Radiomatic). Glyphosate was analyzed by ionic chromatography on a Sax Adsorbosphere column (5 μm , 250×4.6 mm, Alltech), the mobile phase being KH_2PO_4 0.34 g L^{-1} adjusted to pH 2.1 with H_3PO_4 . Mobile phase flow was set at 1.0 mL min^{-1} , and the injected sample volume varied between 100 and 500 μL depending on radioactivity content.

Radioactivity in the soil samples containing the non-extractable residues was measured by liquid scintillation counting of the $^{14}\text{CO}_2$ evolved after combustion in triplicate of 150 mg of ground dry soils using a Sample Oxidizer 307 (Packard Instruments). The entire leaves were fully combusted.

Finally, the carbon content of GT and non-GT leaves was determined using a total organic carbon analyzer (Shimadzu TOC-5050A).

2.6. Data analysis

Herbicide mineralization kinetics were fitted with first-order kinetics according to the following equation:

$$C_t = C_{\max}(1 - \exp(-kt)) \quad (1)$$

where C_t is the amount of mineralized herbicide (% of initial applied dose) at time t (day), $C_{\max}$ is the maximum mineralized (% of initial applied dose), and k the first-order rate constant of mineralization (day^{-1}). The values of $C_{\max}$ and k were determined by non-linear regression (Marquardt-Levenberg algorithm, SigmaPlot, SPSS Inc.). Mineralization half-life ($DT50$) was calculated as:

$$DT50 = \ln 2/k \quad (2)$$

Significant differences between the mineralization kinetics of glyphosate for all conditions of incubation (control, type of plant, washing or no washing, fragmentation of the leaves, placement in soil) were determined with an analysis of variance (ANOVA). The homogeneity of variances was checked with the test of Levene. Pairwise comparisons were done with the test of Wilcoxon. Significant differences between the overall balances of glyphosate fate were determined with the test of Kruskal-Wallis, after which pairwise comparisons were done with the Tukey test. All statistical analyses were performed at the significance level $P = 0.05$ with the R software (R Development Core Team, 2015).

3. Results and discussion

Radioactivity recoveries ranged from 83.1 ± 13.2 to $94.6 \pm 5.3\%$ of the applied radioactivity (AR) for washed GT oilseed rape, from 88.1 ± 0.8 to $93.2 \pm 0.6\%$ AR for non-washed GT oilseed rape, from 88.3 ± 9.2 to $105.1 \pm 13.9\%$ AR for washed non-GT oilseed rape, from 85.5 ± 4.7 to $93.9 \pm 0.7\%$ AR for non-washed non-GT oilseed rape, and were $94.6 \pm 1.6\%$ AR for control. The recoveries lower than 90% were always obtained for the crushed leaves (without bags) because of loss of ^{14}C following crushing of the leaves in the glass mortar. All other recovery rates were satisfactory and fell in the 90–110% range, as recommended by OECD (2002).

3.1. Modification of the soil fate of glyphosate when contained into plant residues

The soil fate of glyphosate contained in plant residues was significantly different from that of glyphosate directly applied to

the soil. The mineralization rate of glyphosate decreased, and its half-life increased almost four-fold (Table 2). The fraction of extractable and non-extractable residues increased up to three and two times, respectively, compared to the control (Fig. 1). After 80 days, the fraction of glyphosate mineralized ranged from 51.6 ± 3.6 to $77.3 \pm 1.0\%$ AR for all samples containing glyphosate in plants, which was significantly lower ($P < 0.001$) than the AR of glyphosate mineralized in control samples (amounting to $82.1 \pm 1.8\%$ AR) (Figs. 1 and 2). The trapping of glyphosate in plant materials probably acted as a protection against mineralization, and in any case resulted in a significant increase in the amounts of total extractable ^{14}C ($P < 0.001$) (Figs. 1 and 3). The latter ranged from 5.9 ± 0.5 to $13 \pm 7.7\%$ AR when glyphosate was returned to the soil with crop residues, against $4.3 \pm 0.1\%$ AR for the control. This is consistent with the results of Doublet et al. (2009), but not with those of Von Wieren-Lehr et al. (1997) who did not observe any modifications after soil incubation of soybean cells associated to glyphosate. In any case, the ^{14}C in the soil extracts (CaCl_2 and NH_4OH) mostly corresponded to the main metabolite of glyphosate, aminomethylphosphonic acid (AMPA), as frequently observed in studies focused on the degradation of glyphosate in soils (e.g. Mamy et al., 2005). However, plant residues apparently protected glyphosate from degradation since the latter was still detected in leaves after 80 days of incubation (Fig. 3).

In general, an increase in the amounts of soil NER was observed following the degradation of plant-derived glyphosate compared to the glyphosate applied to the soil ($P < 0.001$). There were still some exceptions occurring with the crushed leaves incubated without bags, the washed GT oilseed rape entire leaf on soil surface, and the washed non-GT oilseed rape crushed leaves in bags (Figs. 1 and 3). Thus, after 80 incubation days, from 8.7 ± 0.5 to $19.1 \pm 0.4\%$ AR for all plant-glyphosate samples were under NER form, while NER made up $8.3 \pm 0.2\%$ AR for the control (Figs. 1 and 3). The formation of NER leads to a decrease in the toxicity and bioavailability of pesticides in the short term, but could lead to further environmental contamination in the longer run (Barriuso et al., 2008).

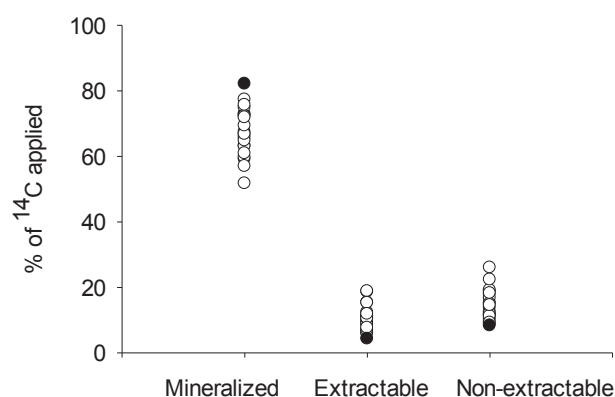


Fig. 1. Summary of the mineralized, extractable, and non-extractable ^{14}C after 80 days for (●) control (glyphosate application on soil), and (○) for all other conditions of incubation (glyphosate residues in glyphosate-tolerant and non-tolerant oilseed rape leaves washed or non-washed, for different fragmentation of leaves and placement in soil).

Thus, this risk will be enhanced if glyphosate is returned to soil within plant residues compared to a direct application on bare soils.

The total soil microbial activity was also monitored during the incubation experiments by measuring total CO_2 . The addition of plant residues increased significantly soil respiration rate from $1.11 \pm 0.04 \text{ mg CO}_2 \text{ kg soil}^{-1}$ for the control to 1.90 ± 0.27 to $2.69 \pm 0.22 \text{ mg CO}_2 \text{ kg soil}^{-1}$ for all incubation samples involving plant residues ($P < 0.001$) (Fig. 4). Thus, the decrease in glyphosate mineralization rates when it was associated with plant leaves cannot be ascribed to an inhibition of leaf decomposition by glyphosate. This is in agreement with Haney et al. (2000) who showed that glyphosate does not inhibit soil microbial activity, and thus the decomposition of plant residues.

As a summary, the soil fate of glyphosate associated to plant tissues was significantly different from the fate of glyphosate in soil: glyphosate mineralization decreased while the amounts of

Table 2

Fractions of mineralized glyphosate residues in glyphosate tolerant (GT) and non-tolerant (non-GT) oilseed rape leaves as a function of leaves fragmentation and placement in soil after 80 days of incubation ($C_{80\text{days}}$), coefficients of mineralization kinetics according to first-order kinetic (maximum amount of mineralized herbicide C_{max} , rate of mineralization k), and half-life of mineralization ($DT50$).

Oilseed rape	Leaf fragmentation	Placement in soil	Observed	Modeled			
			$C_{80\text{days}} (\%)$	$k (\text{day}^{-1})$	$C_{\text{max}} (\%)$	r^2	$DT50 (\text{days})$
Control			82.1 ± 1.7	0.097 ± 0.011	77.2 ± 2.3	0.98	7.1
GT	Washed, entire	Surface	51.6 ± 3.6	0.029 ± 0.002	57.3 ± 1.6	0.99	23.9
	Washed, entire	Middle	63.1 ± 0.2	0.053 ± 0.004	61.8 ± 1.8	0.99	13.1
	Washed, 3 mm	Homogeneous	65.6 ± 2.1	0.047 ± 0.002	65.6 ± 1.3	0.99	14.9
	Washed, crushed	Homogeneous	69.9 ± 0.3	0.055 ± 0.003	68.6 ± 1.5	0.99	12.6
	Washed, crushed, 20 μm bag	Middle	59.6 ± 4.4	0.046 ± 0.002	59.6 ± 1.3	0.99	15.2
	Non-washed, entire	Surface	59.1 ± 2.8	0.026 ± 0.004	68.6 ± 5.9	0.98	26.2
	Non-washed, entire	Middle	66.6 ± 4.7	0.046 ± 0.004	68.1 ± 2.4	0.99	15.1
	Non-washed, 3 mm	Homogeneous	67.4 ± 1.8	0.045 ± 0.003	68.7 ± 1.9	0.99	15.2
	Non-washed, crushed	Homogeneous	72.9 ± 0.7	0.074 ± 0.005	70.3 ± 1.4	0.99	9.4
	Non-washed, crushed, 20 μm bag	Middle	56.9 ± 2.9	0.027 ± 0.002	64.5 ± 1.8	0.99	25.7
	Washed, entire	Surface	72.5 ± 4.4	0.032 ± 0.005	79.7 ± 5.3	0.98	21.5
	Washed, entire	Middle	74.9 ± 2.9	0.053 ± 0.006	75.3 ± 3.0	0.98	13.1
Non-GT	Washed, 3 mm	Homogeneous	72.2 ± 0.5	0.055 ± 0.006	72.2 ± 2.7	0.98	12.5
	Washed, crushed	Homogeneous	66.7 ± 7.4	0.055 ± 0.004	65.9 ± 1.5	0.99	12.5
	Washed, crushed, 20 μm bag	Middle	60.9 ± 7.2	0.033 ± 0.003	65.4 ± 2.8	0.99	20.7
	Non-washed, entire	Surface	64.7 ± 8.8	0.031 ± 0.005	72.4 ± 5.6	0.98	22.7
	Non-washed, entire	Middle	77.3 ± 1.0	0.062 ± 0.006	76.9 ± 2.6	0.98	11.2
	Non-washed, 3 mm	Homogeneous	75.6 ± 1.6	0.069 ± 0.005	74.3 ± 1.8	0.99	10.0
	Non-washed, crushed	Homogeneous	69.3 ± 2.7	0.068 ± 0.006	67.7 ± 1.8	0.99	10.2
	Non-washed, crushed, 20 μm bag	Middle	71.8 ± 2.4	0.042 ± 0.005	74.8 ± 3.6	0.98	16.5

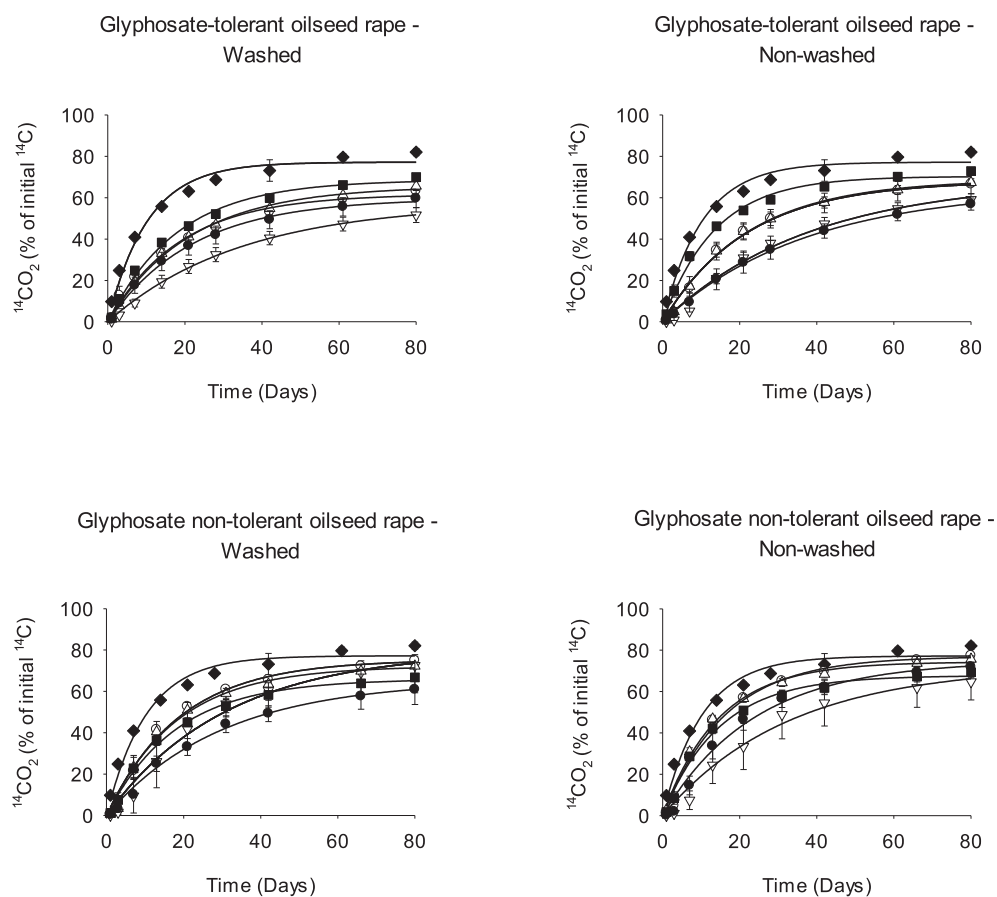


Fig. 2. Mineralization kinetics of glyphosate residues in glyphosate-tolerant and non-tolerant oilseed rape leaves, washed or non-washed, as a function of the fragmentation of leaves and of their placement in soil: (◆) control; (■) crushed, homogeneous; (△) 3-mm size, homogeneous; (○) entire, middle; (●) crushed, bag; (▽) entire, surface (standard deviations are shown only when larger than symbols). Lines are adjustment with first-order kinetics.

extractable ^{14}C and NER increased, leading to an increase in glyphosate persistence in the environment.

3.2. Crop residues management changes the soil fate of glyphosate contained into plant residues

Glyphosate is largely used in conventional tillage and no-tillage agricultural systems (Cerdeira and Duke, 2006; Dorn et al., 2013; Locke et al., 2008). However, the management of crop residues varies according to these systems: fragmentation and burying in tillage systems, leaving crop residues on the soil surface in no-tillage systems (Beare et al., 1993; Guérif et al., 2001). The results of the different incubation conditions showed that, in general, the mineralization rates of glyphosate decreased significantly as the size of plant residues increased, and as the homogeneity of the soil/leaf mixture decreased ($P < 0.05$) (Fig. 2, Table 2). The lowest mineralization rates were always found for the entire leaf left on soil surface and for the crushed leaf in bag, and the highest for the crushed leaf and the 3-mm size fragments of leaf (Fig. 2, Table 2). The mineralization rates of glyphosate in oilseed rape leaves were 1.3 times (for crushed leaves homogeneously mixed in the soil) to 3.7 times (for entire leaves left on soil surface) lower than the mineralization rate of the control (Table 2). This may be attributed to a lower decomposition rate for larger plant residues compared to crushed leaves, entailing a lower availability of associated glyphosate for microorganisms (Angers and Recous, 1997; Beare et al., 1993; Coppens et al., 2006; Iqbal et al., 2014; Sørensen et al.,

1996). The size and placement of the plant residues mostly had no significant effect on the total soil microbial activity monitored by measurements of total CO_2 evolved ($P > 0.05$) (Fig. 4).

Glyphosate mineralization showed a 3-days lag phase for those samples involving entire leaves, compared to the other treatments. This was particularly true when the leaves were located on the soil surface, and for the leaves in polyamide bags (Fig. 2). This lag phase may be attributed to a lower availability of glyphosate in a less decomposed plant material and in bags (Benoit et al., 1999). The crushed leaves in polyamide bags were not directly in contact with the soil, but some ^{14}C was nevertheless extracted from the soil and the amounts were significantly higher than the control ($P < 0.001$) (Fig. 3). This indicates that there was a transfer of radioactivity from plant material to the soil probably due to microbial activity.

As could be expected from the mineralization rate patterns across treatments (Table 2), the lowest amounts of extractable ^{14}C occurred with the crushed leaves mixed in the soil, while the highest amounts occurred for the entire leaves left on the soil surface (Fig. 3).

The highest amounts of NER were found for the entire leaves, whether located on the soil surface or within the soil, and the lowest amounts for the crushed leaves homogeneously mixed into the soil. As indicated in section 3.1, there were no significant differences between the amounts of NER in soils mixed with crushed leaves and those of the control ($P > 0.05$). This is consistent with the higher mineralization rates of glyphosate under this condition (Table 2) (Barriuso et al., 2008).

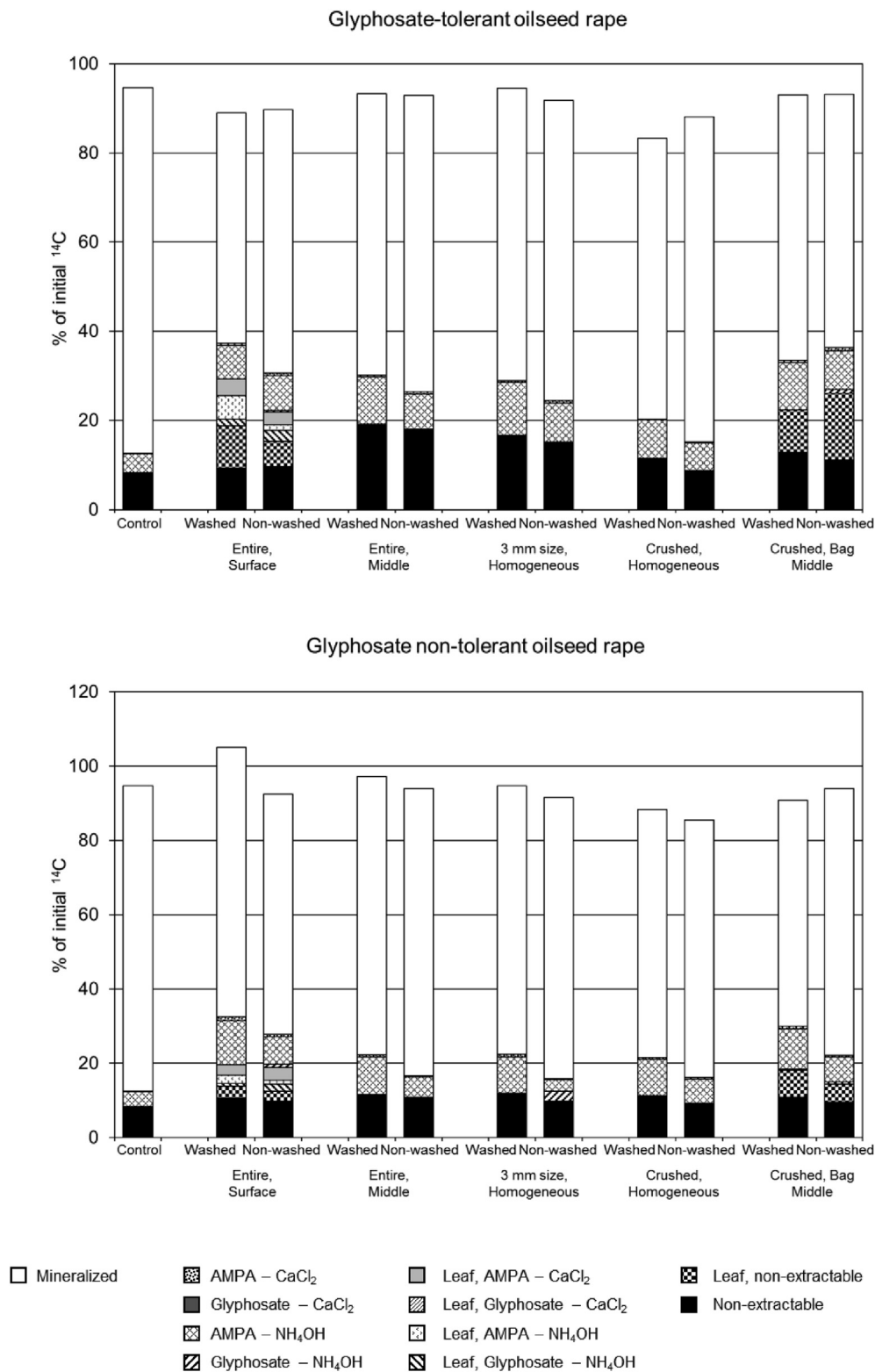


Fig. 3. Fate in soil of glyphosate residues in glyphosate-tolerant and non-tolerant oilseed rape leaves, washed or non-washed, as a function of the fragmentation of leaves and of their placement in soil after 80 days.

Overall, the size of plant residues and their placement in soil had a strong impact on the soil fate of glyphosate associated to plant material: the persistence of glyphosate increased as the size of plant residues increased and their distribution in soil was less

homogeneous. Thus, in no-tillage agricultural systems where plant residues are left on soil surface, an increase in the soil persistence of pesticides trapped in plant material may be expected.

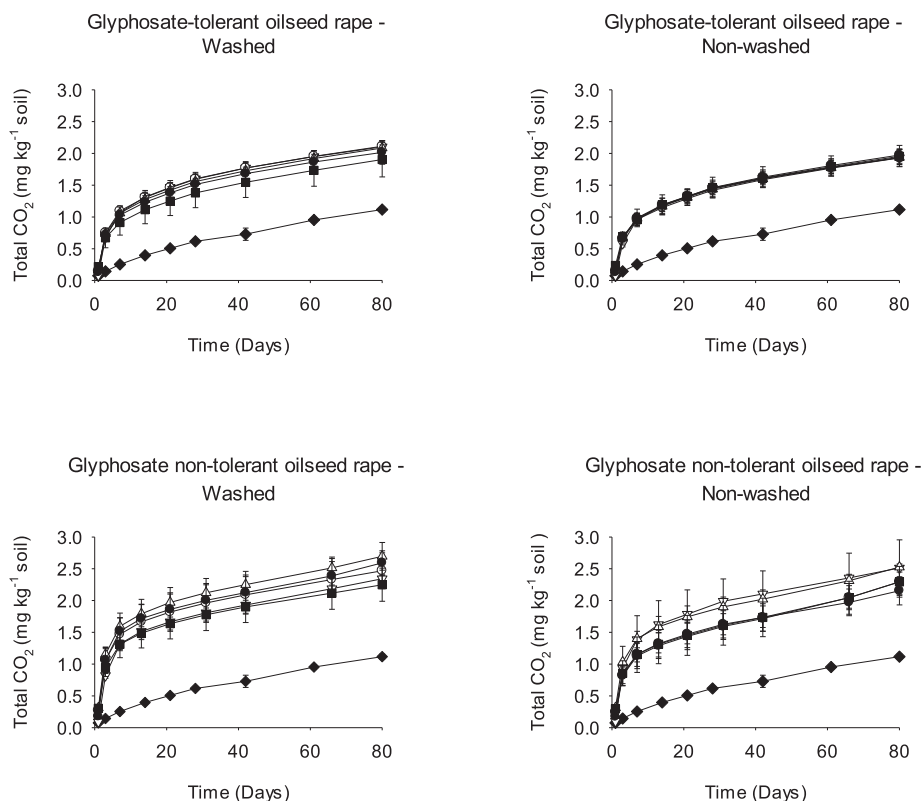


Fig. 4. Mineralization kinetics of total CO_2 in glyphosate-tolerant and non-tolerant oilseed rape leaves, washed or non-washed, as a function of the fragmentation of leaves and of their placement in soil: (◆) control; (■) crushed, homogeneous; (△) 3-mm size, homogeneous; (○) entire, middle; (●) crushed, bag; (▽) entire, surface (standard deviations are shown only when larger than symbols).

3.3. Effect of the interaction degree between plant tissues and glyphosate

Glyphosate intercepted by leaves is first adsorbed on the cuticle prior to diffusing into leaf organic matter and being absorbed by leaves. Desorption with a water solution made it possible to estimate the amounts of glyphosate adsorbed, while the absorbed fraction could be taken as the amount of glyphosate remaining in the plant residues after water desorption. Foliar absorption of glyphosate in washed leaves corresponded to $27.0 \pm 10.6\%$ AR in GT oilseed rape, and to $30.6 \pm 14.9\%$ AR in non-GT oilseed rape. These amounts are consistent with those reported by Doublet et al. (2009), but they are lower than those evidenced by Nandula et al. (1999). This may be due to the latter using glyphosate solutions containing surfactants, as opposed to pure water solutions in our case. There were no significant differences in glyphosate absorption between GT and non-GT oilseed rape ($P > 0.05$).

In general, for both oilseed rape lines, the mineralization of glyphosate was higher for unwashed leaves than for washed leaves ($P < 0.05$) (Fig. 2, Table 2). This probably arose because non-absorbed glyphosate was more directly available to soil microorganisms, and/or because of the higher amounts of glyphosate present in the unwashed material, which stimulated soil microbial activity (Doublet et al., 2009; Haney et al., 2000). The extractable amounts were lower with unwashed leaves than with washed leaves ($P < 0.001$) (Fig. 3, Table 2). As mentioned above for mineralization, these differences could indicate that the glyphosate present on the leaf surface enhances microbial activity, thereby accelerating leaf decomposition and making absorbed glyphosate more accessible to soil microorganisms.

Entire leaves left on soil surfaces were extracted separately to

determine their glyphosate and AMPA contents after 80 days (see section 2.4. Incubation procedure). The total extractable ^{14}C from leaves was $10.4 \pm 2.4\%$ AR for washed GT leaves, $6.5 \pm 0.4\%$ AR for non-washed GT leaves, while it was $5.8 \pm 2.5\%$ AR for washed non-GT leaves, and $6.3 \pm 7.0\%$ AR for non-washed non-GT leaves. There were no significant differences among all four treatments ($P > 0.05$). These amounts are lower than those observed by Rampoldi et al. (2011) for other species like maize and soybean, however in the latter experiments crop residues were incubated in the absence of soils. After 80 days, CaCl_2 extracts only contained AMPA, which is consistent with the degradation pattern of glyphosate in GT oilseed rape (Nandula et al., 2007). However, NH_4OH extracts contained, in addition to AMPA, significant amounts of glyphosate: from $0.8 \pm 0.2\%$ AR for washed non-GT leaves to $2.6 \pm 0.2\%$ AR for non-washed GT leaves. The absorption of glyphosate in plant material therefore provided a protection against soil degradation. The formation of soil NER from glyphosate in entire leaves on soil surface ranged from 9.2 ± 2.5 to $10.5 \pm 2.4\%$ AR, without any significant differences between washed and non-washed leaf samples ($P > 0.05$).

It should be noted that this study was based on pure glyphosate, not on commercial products containing surfactants aiming at increasing the absorption of glyphosate in plants (Doublet et al., 2009). Commercial formulations are thus likely to increase the amounts of glyphosate returning to soils via plant residues. The soil fate of glyphosate contained in these residues will also depend on plant species, as the biochemical composition of crops will impact their decay (Johnson et al., 2007), on the absorption, translocation, and degradation of the herbicide in the treated plant (Gauvrit, 1996; Grangeot et al., 2006; Locke et al., 2008), and on the plant compartment glyphosate ends up in (Doublet et al., 2009).

When glyphosate was absorbed in plant, entailing a high degree of interactions between the molecule and plant tissue, its degradation in soil decreased and glyphosate was still detected more than 80 days after its application. In similar incubation conditions (Dijon soil, 28 °C, in darkness), but where glyphosate was directly applied to the soil, it was not detected after 30 days (Mamy et al., 2005). It has to be underlined that these experiments were performed at an optimal temperature of 28 °C so that in field conditions, where temperatures fluctuate and are generally lower than 28 °C, the soil persistence of glyphosate associated to plant material will largely increase.

3.4. The fate of glyphosate contained into plant residues might depend on the tolerant-glyphosate character of oilseed rape

The mineralization of glyphosate in GT oilseed rape was lower than that of glyphosate in non-GT oilseed rape, except when the leaves were crushed and mixed into the soil ($P < 0.05$) (Fig. 2, Table 2). After 80 days, the fraction of glyphosate mineralized in soil ranged from 51.7 ± 3.6 to $69.9 \pm 0.3\%$ AR for washed GT oilseed rape, and from 60.8 ± 7.2 to $74.9 \pm 2.9\%$ AR for washed non-GT oilseed rape.

Soil microbial activity was lower with the GT oilseed rape compared to the non-GT one ($P < 0.01$) (Fig. 4). This can be explained by the lower carbon content of GT oilseed rape leaves compared to non-GT oilseed rape leaves: the latter contained $40.0 \pm 2.6\%$ C and the former $44.5 \pm 0.5\%$. These differences should not be attributed to genetic modifications since both varieties were not obtained from the same line.

There were no significant differences in the glyphosate extractable amounts between the two lines of oilseed rape for washed leaves ($P > 0.05$), but significant differences ($P < 0.001$) occurred for unwashed leaves (except for the entire leaf at soil surface and for the crushed leaf mixed in soil): the extractable amounts were higher for GT oilseed rape than for non-GT oilseed rape (Fig. 3). This is consistent with the lower mineralization rates of glyphosate residues in GT oilseed rape leaves compared to non-GT leaves (Table 2).

In general, the NER were significantly different between GT and non-GT oilseed rape ($P < 0.05$). Thus, for example, the NER from entire oilseed rape leaves made up $9.6 \pm 3.9\%$ of AR for washed GT oilseed rape, against $3.3 \pm 0.4\%$ AR for washed non-GT oilseed rape (Fig. 3). These fractions are also lower than those observed by Rampoldi et al. (2011) for maize and soybean. The incubation of plant-glyphosate residues into the polyamide bags, without direct contact with soil, showed the formation of soil NER, and the amounts were higher than the control ($P < 0.001$, except for the non-washed non-GT oilseed rape) (Fig. 3): NER made up $9.5 \pm 0.7\%$ AR for non-washed non-GT crop and $12.9 \pm 0.9\%$ of AR for washed GT crop. Similarly to extractable ^{14}C , this reveals a transfer of radioactivity from leaves to soils, probably due to microbial activity (Barriuso et al., 2008).

In general, the soil fate of glyphosate associated to GT oilseed rape was different from that of glyphosate associated to non-GT oilseed rape. Thus the GT trait of this crop might play a direct or indirect role in the soil fate of glyphosate in plant residues.

4. Conclusion

The soil fate of glyphosate contained in plant residues was significantly different from that of glyphosate directly applied to soil. The mineralization of glyphosate decreased, resulting in a half-life increasing almost four-fold compared to the control, and the fraction of extractable and non-extractable residues increased up to three and two times, respectively. The persistence of glyphosate in

soil increased as the size of plant residues increased and as their distribution in soil was less homogeneous. In general, the amounts of glyphosate remaining after 80 days were low, soil extractable amounts being mainly constituted of one metabolite, AMPA. However, glyphosate was still detected after 80 days when an entire leaf was left on the soil surface. Glyphosate absorption in plant is a protection against its degradation by soil microorganisms, which increases its persistence in the environment. A significant transfer of radioactivity from leaf to soil was also evidenced, making up more than 20% of the applied rate. Finally, the results showed that the degradation of glyphosate in GT oilseed rape could be slower than with a crucifer, non-GT weed. The increase in extractable glyphosate and AMPA following soil degradation of glyphosate contained in oilseed rape leaves could increase the potential risk of groundwater contamination by these molecules. Likewise, the increase in NER formation may entail further, albeit delayed contamination of the environment.

Plants might contribute to increase the amounts of glyphosate in soils following crop residue returns or leaf senescence. This trend is more pronounced with GT crops which accumulate non-degraded glyphosate in their tissues. In addition, GT crops will most likely generate more residual biomass than weeds. When bound with plant material, glyphosate is more persistent in the environment, therefore the risk of environmental contamination by this herbicide, and also by its metabolite AMPA, increases compared to the risk related to a direct application of glyphosate to bare soils. In general, the modifications of the fate in soil of pesticides due to interception by foliage should be taken into consideration in risk assessment and registration procedures.

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